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From Conventional NLP to Large Language Models

What changes and what remains the same?

Examples from pharmacoepidemiology

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What is NLP?

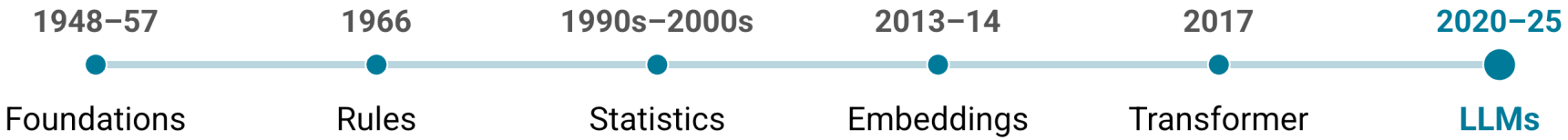
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NLP and large language models

Natural language processing (NLP): computer methods for language.
Large language model (LLM): a model trained on vast text collections.

Example note	“Started Drug A. Felt dizzy two days later.”
NLP task	Extract the medicine, symptom and timing.
Study question	Did Drug A increase the risk of dizziness?

A timeline of NLP and large language models



Text processing → **Learned patterns** → **Flexible tasks**

Earlier methods still have a place in research.

The foundations

- **1948** Shannon
  Language has statistical patterns
- **1950** Turing
  Conversation as a test
- **1954** Georgetown–IBM
  A machine translation demonstration
- **1957** Chomsky
  Formal descriptions of grammar



1960s–1970s

Early NLP followed rules written by people

PERSON

```
I am worried about my exam.
```

ELIZA

```
Why are you worried about your exam?
```

ELIZA (1966) used written patterns to produce a conversational response.

Bag of words and TF–IDF

Three example reports, reduced to their words

REPORT A

drug rash

REPORT B

drug pain

REPORT C

drug cough

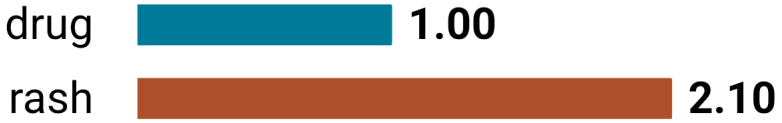
Bag of words (BoW)

Count words in Report A



TF–IDF

Weight counts by distinctiveness



“drug”: 3 of 3 reports. “rash”: 1 of 3. “rash” gets more relative weight.

TF = term frequency. IDF = inverse document frequency.

Text processing: keeping “No”

Clean No bleeding. → **No bleeding.**

Split



Encode 12 41 7

“No bleeding” → **NEGATED**

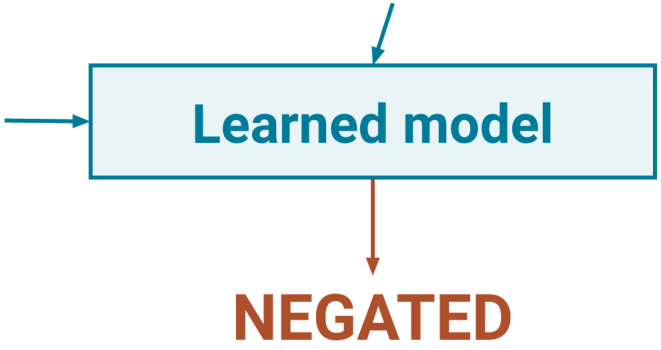
Statistical NLP: learning from examples

People label examples

Dizziness after Drug A.	REPORTED
No dizziness after Drug A.	NEGATED
Rash after Drug B.	REPORTED
Denies rash after Drug B.	NEGATED

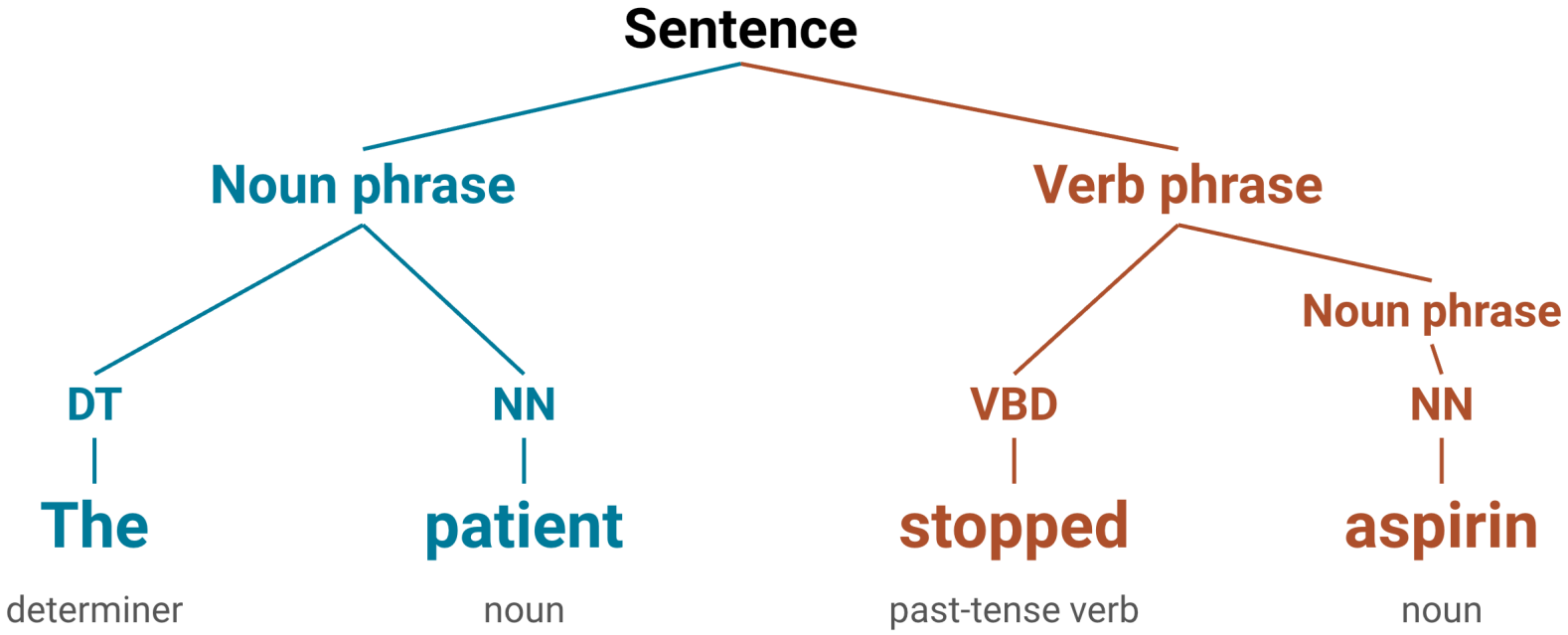
New report

No rash after Drug C.



A new medicine name, the same negation cue

Penn Treebank: labels and sentence structure



Medication extraction: shared tests

2009 i2b2

Drug A	**10 mg**	**daily**	for	**pain**
Medicine	Dose	Frequency		Reason

2018 n2c2

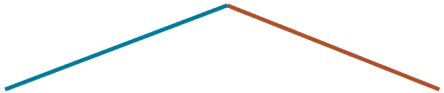
Rash attributed to **Drug B.**

Event — Recorded link — Medicine

Learning from text itself

Topic models

"Costly refill. Itchy rash."



refill
cost
pharmacy

rash
itch
hives

Access

Symptoms

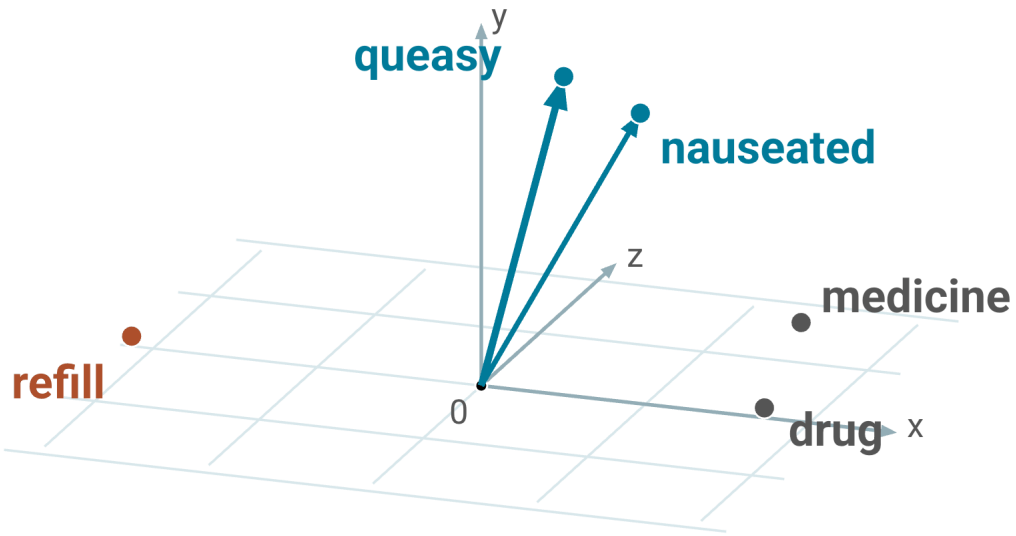
Early neural language models

The patient took the



Predict the next word

Word2Vec and the word-embedding space



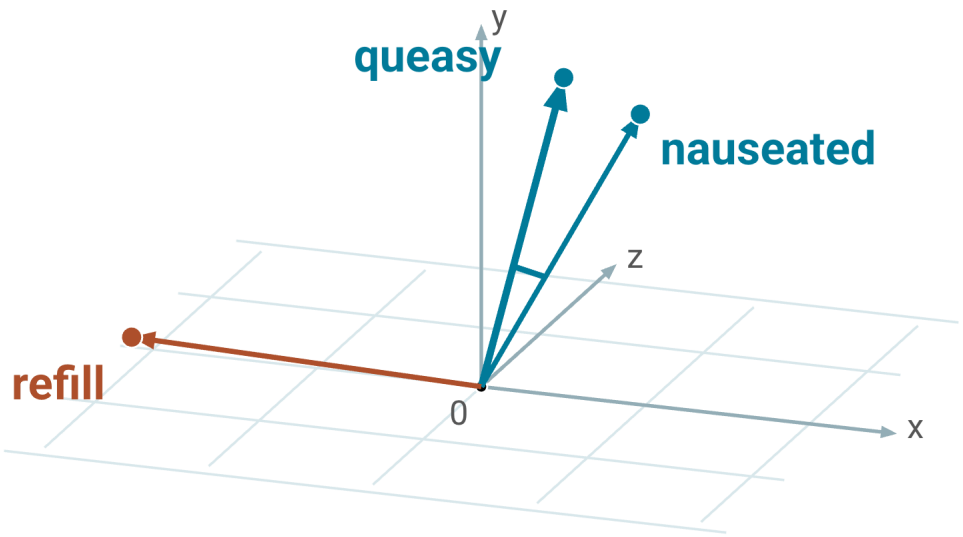
Word2Vec (2013)

Learns word vectors by predicting nearby words.

“felt queasy after …”
“felt nauseated after …”

Related contexts can yield similar vector directions.

Vector directions and language similarity



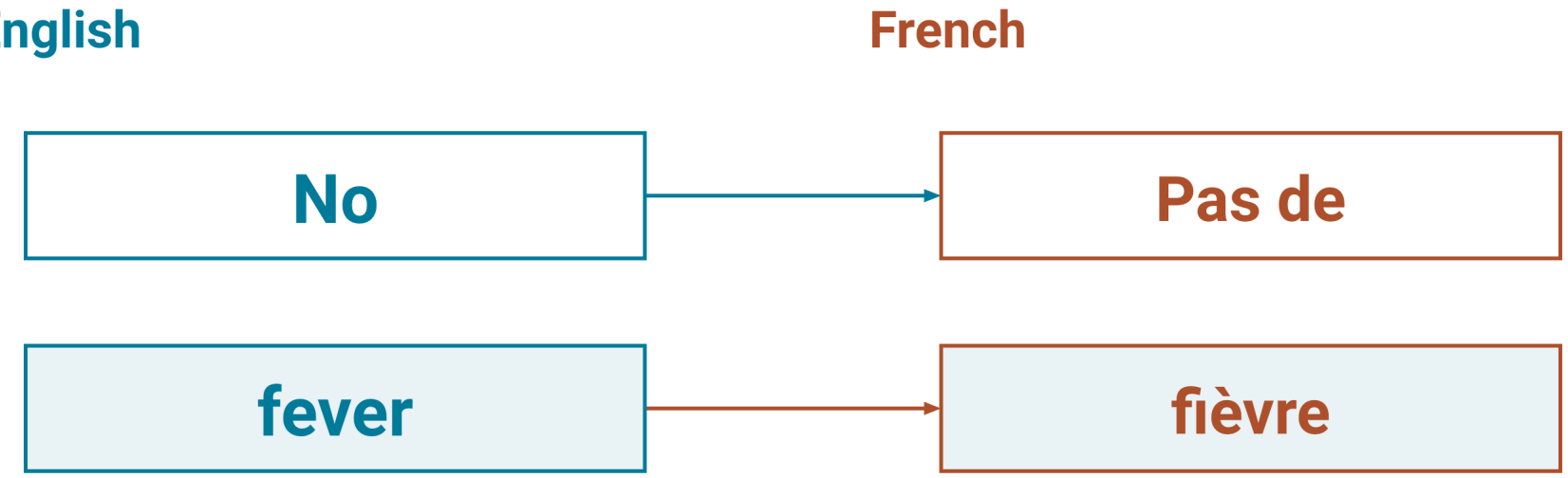
Direction helps compare words

Each arrow is a vector:
a list of numerical coordinates.

A smaller angle means greater cosine similarity.

Search “queasy”
Find related wording: “nauseated”

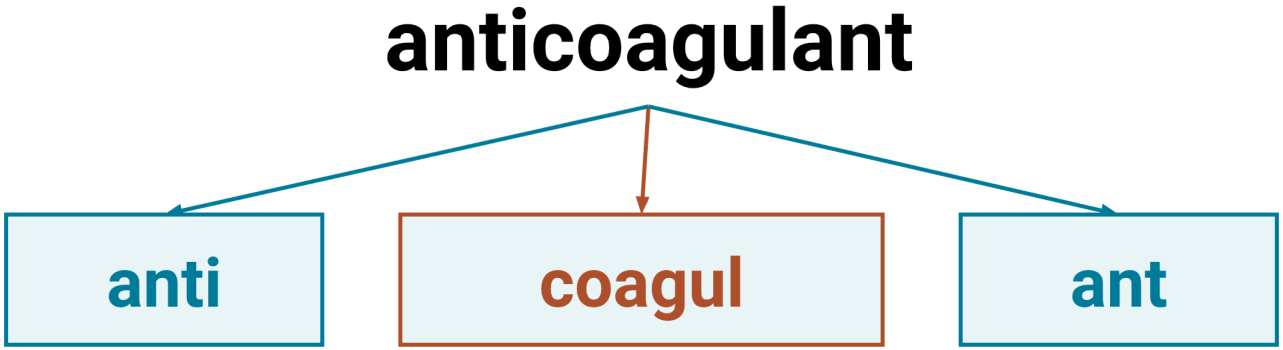
Neural translation and attention



Attention uses the relevant input while writing.

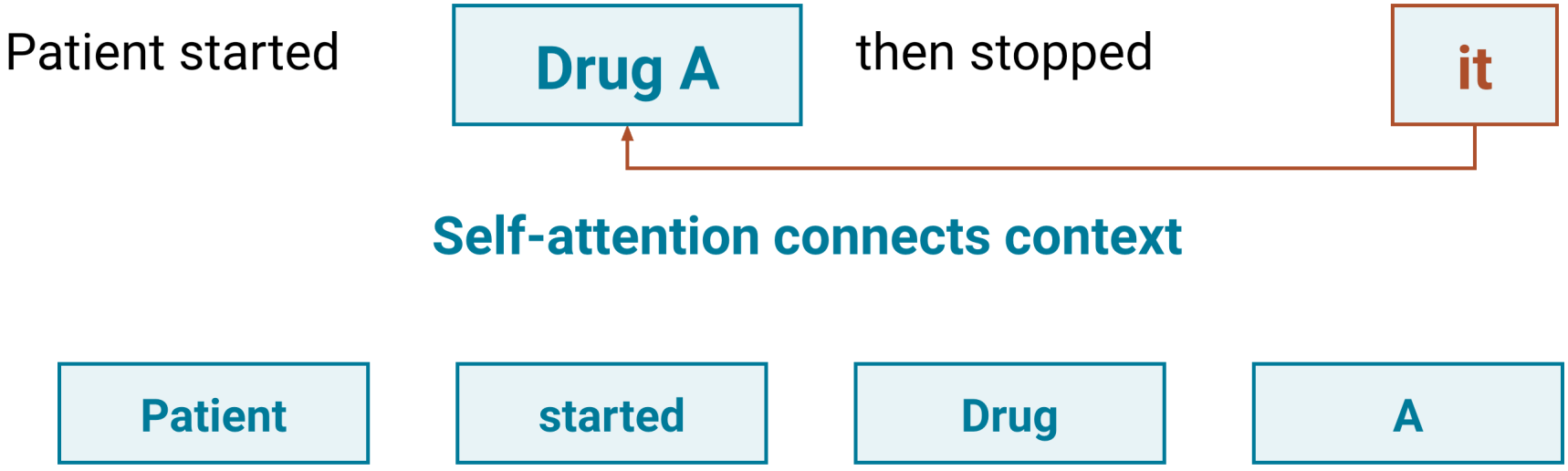
Tokens: pieces of text

A token is a piece of text



A rare word can use familiar pieces.

Transformers: connecting the context



Context changes a word’s representation

The same word, different contexts

No bleeding today.

Current: negated

Bleeding **last year**.

Historical mention

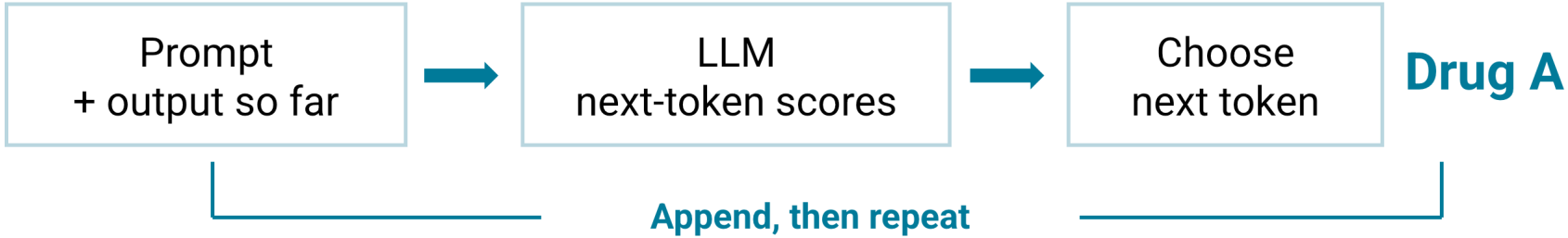
Contextual embeddings change with the sentence.

How an LLM generates text

Instruction: Extract the medicine name.

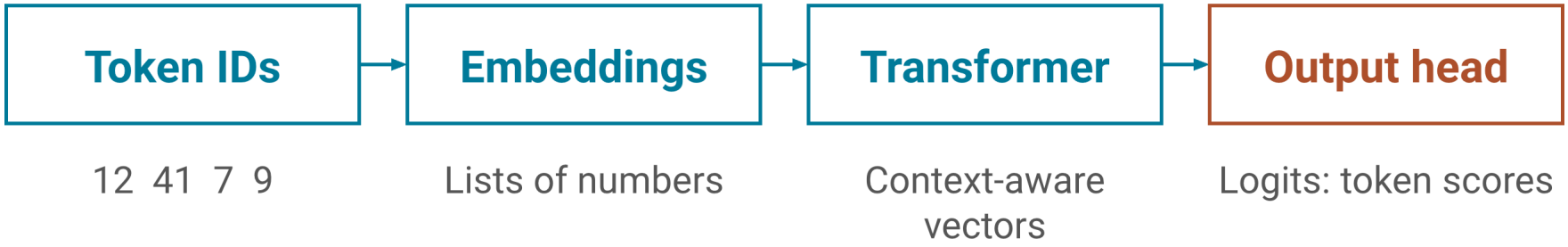
Note: “Stopped Drug A yesterday.”

Stop when the answer is complete



Inside a text-generating LLM

“Stopped Drug A.”



Softmax Scores become probabilities

Parameters (weights) are the model’s learned numbers.

Attention: queries, keys and values

Drug A … then stopped it.

Q Query	→ K Key	→ V Value
What fits here?	What can match?	What information to pass?
“it”	“Drug A”	Medicine context

Attention matches Q with K, then mixes V information.

Training, fine-tuning and inference

Pre-training

"The patient took …"

medicine

Weights change

Fine-tuning

Extract: "Drug A stopped."

Drug A

Weights change

Inference

"Stopped Drug B."

Drug B

Weights stay fixed

How training updates the weights

“The patient took the …”

Target: medicine



Loss

Prediction error

Compare with “medicine”

Backpropagation

Work out weight updates

After an illustrative update: higher probability for the target

Learning useful responses

Summarise: “Dizzy after Drug A. Unsure if related.”

Supervised fine-tuning (SFT)

Learn from good examples

“Dizziness reported. Relationship uncertain.”

RLHF

Reinforcement learning from human feedback

Prefer this wording over “Drug A caused dizziness.”

Learning a task from the prompt

Examples inside the prompt

No dizziness with Drug A. **NEGATED**

Dizziness after Drug A. **REPORTED**

Felt dizzy after Drug B. **REPORTED**

Few-shot prompting: examples guide the answer.

The context window

Context window = the available token space



Prompt + supplied records

Output

A longer prompt leaves less room for the answer.

max_tokens: the output budget

Tokens fill the available spaces

max_tokens = 3



Limit reached

max_tokens = 8

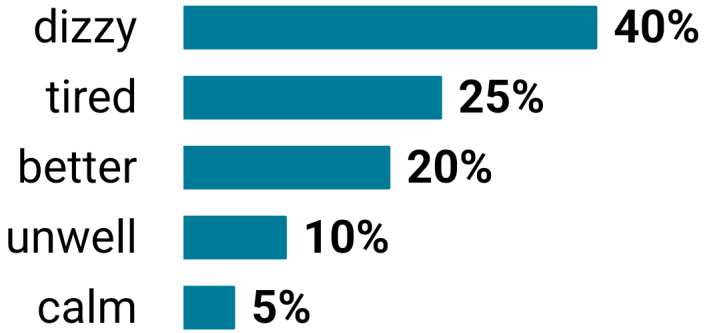


Answer finished

Temperature, top_k and top_p

“The patient felt …”

temperature = 1.0



temperature

Change the spread

top_k

Choice count

top_p

Combined probability

One prompt at two temperatures

Summarise in one short sentence:

"Started Drug A on Monday. Felt dizzy on Wednesday. Stopped it on Thursday."

Low temperature 0.2

I started Drug A on Monday, **felt dizzy on Wednesday**, and **stopped it** on Thursday.

High temperature 1.2

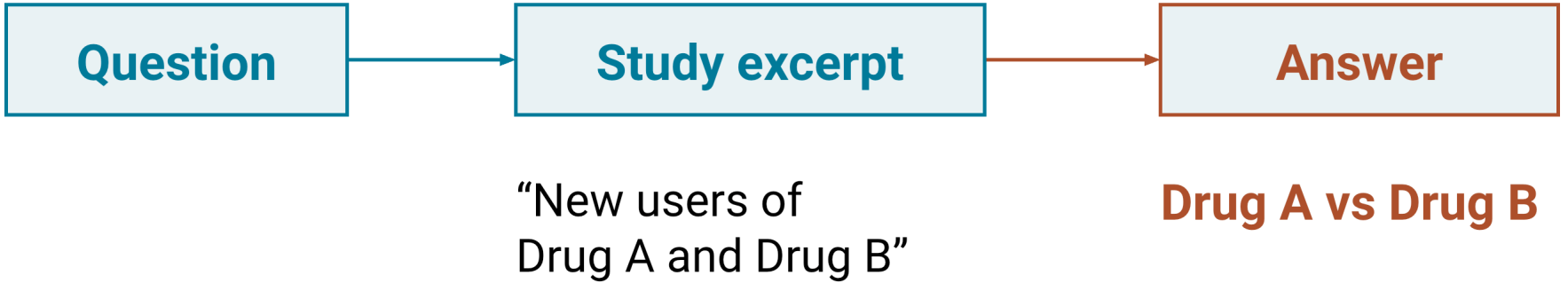
I started Drug A on Monday, **experienced dizziness by Wednesday**, and **stopped taking it** on Thursday.

Run 3 of 3

Recorded responses: Qwen3 4B

Retrieval-augmented generation (RAG)

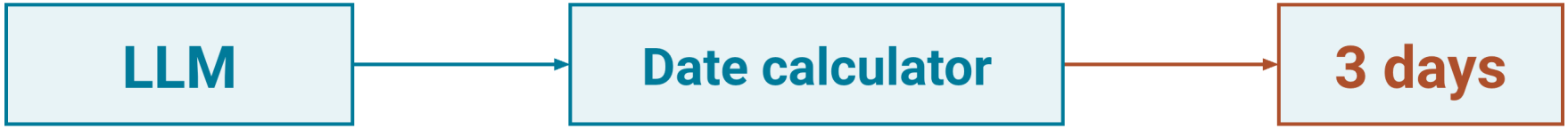
Which treatments did the study compare?



RAG: retrieve a source, then use it to answer.

Tool use: a calculator example

Onset: 5 May. Treatment started: 2 May.



5 May − 2 May

A tool call asks another program to perform a step.

Reasoning: checking a sequence of events

“Started 2 May. Dizziness 5 May. Stopped 6 May.”



Check the dates, then answer: onset after 3 days.

WHAT CHANGED

The same extraction task across the eras

“The patient stopped Drug A yesterday.”

Rules	Statistical NLP	Pretrained model	Instruction LLM
Dictionary match	Labelled examples	Task fine-tuning	“Which medicine?”
Drug A	**Drug A**	**Drug A**	**Drug A**

From narrative text to research data

Text says	Extracted information	Research use
“Drug A prescribed”	Medicine: Drug A Status: prescribed	Identify treatment records
“Drug A stopped”	Medicine: Drug A Status: stopped	Describe treatment changes
“Rash after Drug A”	Symptom: rash Sequence: after Drug A	Find reports for event review
“Symptoms improved”	Symptom trend: improved	Describe reported outcomes

WHAT REMAINS THE SAME

The same need for accurate measurement

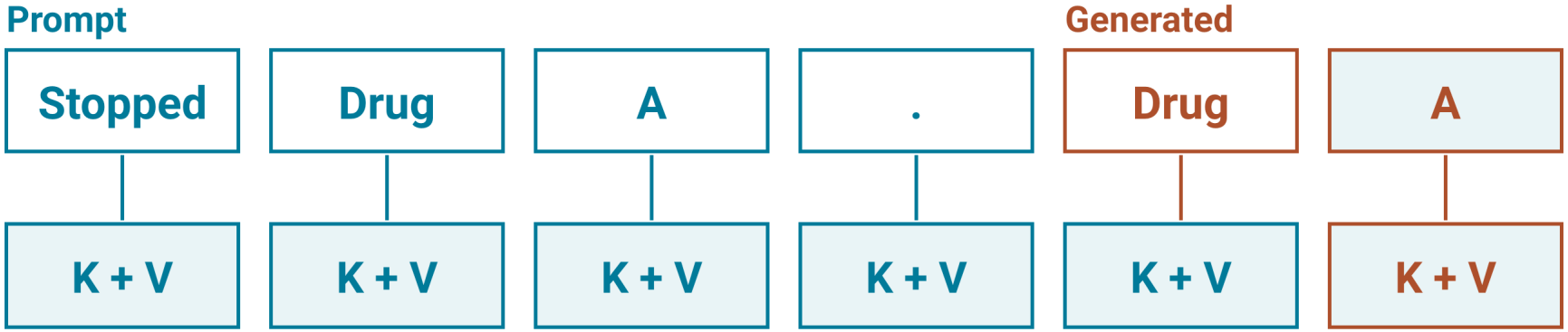
Source note: “No bleeding. Drug A stopped 6 May.”

Event status	**NEGATED**
Stop date	**6 May**
Source record	**Note + model version**

Compare extracted fields with human-reviewed source text.

KV cache: reusing earlier calculations

Decode: generate the next token



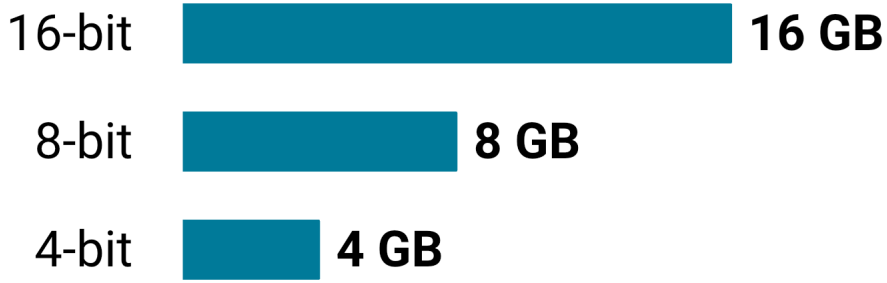
KV cache

Reuse earlier K and V vectors

Less repeated computation, more memory to store the cache.

Memory, GPUs and quantisation

8B parameters: raw weight storage



GPU

Runs parallel calculations

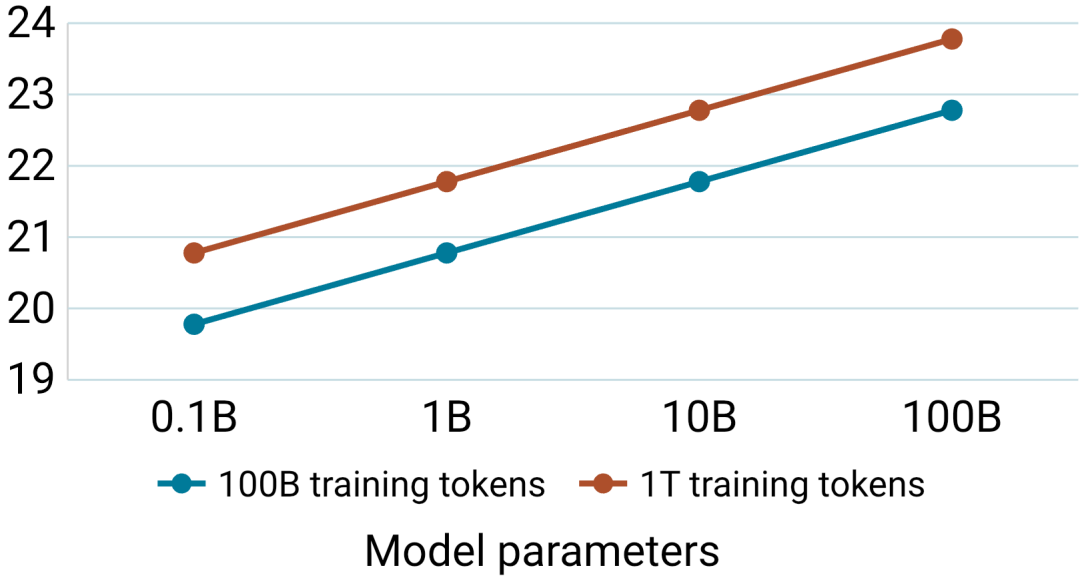
VRAM

GPU working memory

Quantisation: fewer bits per weight. Add cache and working space.

The scale of training computation

Training operations, $\log_{10}$(FLOPs)



Illustrative estimate: 6 × parameters × training tokens

20 to 21

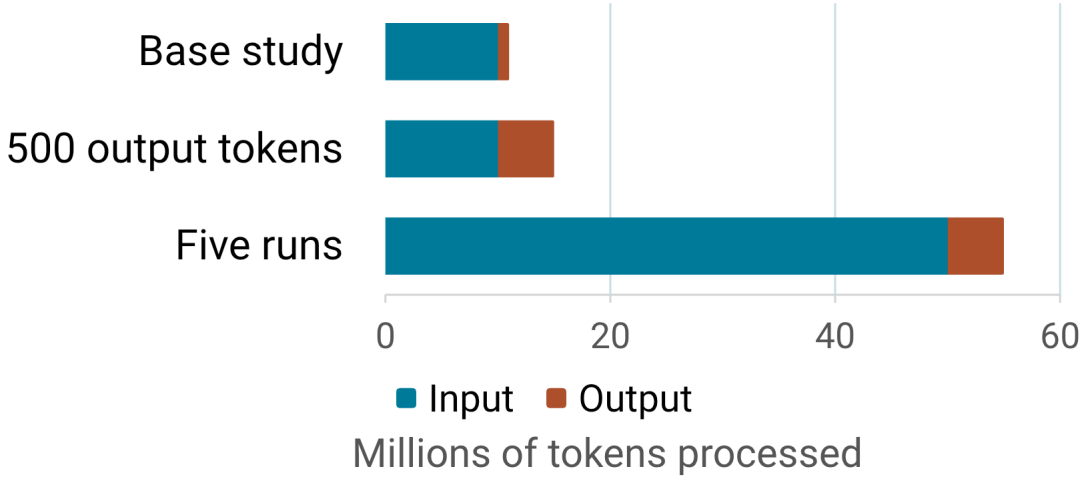
means 10× work

FLOPs: operations
B: billion
T: trillion

The cost of running a study

Example: 10,000 notes

Base: 1,000 input + 100 output tokens per note



Batching

Several notes together

Latency: wait per answer
Throughput: notes per minute

CLOSING

The main changes, and the enduring requirement

Models learned more language patterns.
Instructions made tasks easier to request.

Useful medicine research still needs
accurate measurement and sound study design.

APPENDIX 1

Historical landmarks before 2013

1948–57	Shannon, Turing, translation, grammar
1964	Brown Corpus
1966	ELIZA and the ALPAC report
1968	Regular-expression search
1970–72	SHRDLU
1972–75	IDF and the vector space model
1980	Porter stemming
1985–90	WordNet
1990	Latent semantic analysis
1990s	Statistical tagging and translation
1992–93	TREC and Penn Treebank
1997	LSTM
2001	Conditional random fields
2002	BLEU
2003	LDA, neural LM, phrase-based MT
2011	Watson and neural NLP toolkits

Clinical landmarks: MedLEE (1994), NegEx (2001), i2b2 medication challenge (2009).

APPENDIX 2

Landmarks in neural NLP and LLMs

2013	Word2Vec
2014	GloVe and sequence-to-sequence
2014–15	Attention in neural translation
2016	Subword neural translation
2017	Transformer
2018	ELMo, ULMFiT, GPT and BERT
2019	GPT-2, T5, BioBERT, ClinicalBERT
2020	GPT-3, scaling laws and RAG
2021	FLAN instruction tuning
2022	Chinchilla compute/data balance
2022	Chain-of-thought prompting
2022	InstructGPT and ChatGPT
2022–23	ReAct and tool use
2023	GPT-4 and LLaMA
2024	o1 reasoning models
2025	DeepSeek-R1